Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Meikle Dairy

Physical address of dairy:

Number and Street	City	County	Zip Code
1685 S Hart RD	Modesto	Stanislaus	95358

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 07/01/1950

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X017-X002-X013-XXXX

B. OPERATORS

Meikle, James D Jr

Operator name: Meikle, James D Jr

Telephone no.: (209) 529-2206 Landline, (209) 918-0803 Cellular

Mailing Address Number and Street	City	State	Zip Code
1601 S Hart RD	Modesto	CA	95358

This operator is responsible for paying permit fees.

C. OWNERS

Meikle, James D

Legal owner name: Meikle, James D

Telephone no.: (209) 522-6179 Landline, (209) 484-1665 Cellular

Mailing Address Number and Street	City	State	Zip Code
1685 S Hart RD	Modesto	CA	95358

Meikle, James D Jr

Legal owner name:	Meikle, James D Jr	Telephone no.:	(209) 529-2206	(209) 918-0803
			Landline	Cellular
1601 S Hart RD	Modesto		CA	95358
Mailing Address Number and Street	City		State	Zip Code

This owner is responsible for paying permit fees.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	75	0	125	5
Number under roof	525	75	0	0	0	0
Maximum number	525	75	75	0	125	5
Average number	525	75	75	0	125	5
Avg live weight (lbs)	1,400	1,450	1,000	0		

Predominant milk cow breed: Holstein

Average milk production: 77 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 16,231.55 *tons per reporting period*

Total nitrogen from manure: 209,161.69 *lbs per reporting period*

After ammonia losses (30% loss applied): 146,413.18 *lbs per reporting period*

Total phosphorus from manure: 34,326.25 *lbs per reporting period*

Total potassium from manure: 102,956.82 *lbs per reporting period*

Total salt from manure: 264,442.50 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 10,452,101 *gallons*

Total nitrogen generated: 62,839.26 *lbs*

Total phosphorus generated: 5,805.17 *lbs*

Total potassium generated: 108,786.86 *lbs*

Total salt generated: 503,368.75 *lbs*

	10,452,101	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	10,452,101	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
MID Lateral 5	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
05/24/2023	Solid commercial fertilizer *4-0-0*	18.87 *ton*	Dry-weight	0.1	4.000000	0.000000	0.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	1,508.09	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	1,508.09	0.00	0.00	0.00

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
1	57	57	2	process wastewater	X017-X002-X014-XXXX
2	25	25	2	both	X017-X002-X013-XXXX
3	19	19	2	both	X017-X002-X013-XXXX
9	50	50	2	process wastewater	X017-X001-X002-XXXX
Totals for areas that were used for application	151	151	8		
Totals for areas that were not used for application					
Land application area totals	151	151	8		

B. CROPS AND HARVESTS

1

Field name: 1

10/28/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 57 Plant date: 10/28/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/02/2023	948.23 *ton*	As-is		73.5	5,900.00	1,200.00	9,100.00		9.21

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	17.00	170.00	27.20	141.10	0.00
Total actual harvest content	16.64	196.30	39.93	302.77	812.03

05/24/2023: Corn, silage

Crop: Corn, silage Acres planted: 57 Plant date: 05/24/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/16/2023	1,959.13 *ton*	As-is		70.0	2,800.00	800.00	5,000.00		5.94

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	34.37	192.48	54.99	343.71	1,224.97

2

Field name: 2

10/28/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 25 Plant date: 10/28/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/02/2023	415.89 *ton*	As-is		72.2	7,000.00	1,400.00	11,500.00		11.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	17.00	170.00	27.20	141.10	0.00
Total actual harvest content	16.64	232.90	46.58	382.62	1,100.68

05/24/2023: Corn, silage

Crop: Corn, silage Acres planted: 25 Plant date: 05/24/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/16/2023	859.27 *ton*	As-is		69.1	2,400.00	700.00	5,100.00		5.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	34.37	164.98	48.12	350.58	1,210.75

3

Field name: 3

10/28/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 19 Plant date: 10/28/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/02/2023	316.07 *ton*	As-is		71.9	6,600.00	1,300.00	11,300.00		10.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	17.00	170.00	27.20	141.10	0.00
Total actual harvest content	16.64	219.59	43.25	375.96	981.65

3

05/24/2023: Corn, silage

Crop: Corn, silage Acres planted: 19 Plant date: 05/24/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/16/2023	653.04 *ton*	As-is		70.2	3,200.00	700.00	5,300.00		5.93

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	34.37	219.97	48.12	364.33	1,214.75

9

Field name: 9

10/28/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 50 Plant date: 10/28/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/02/2023	831.78 *ton*	As-is		73.0	6,000.00	1,300.00	10,400.00		10.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	17.00	170.00	27.20	141.10	0.00
Total actual harvest content	16.64	199.63	43.25	346.02	961.20

05/24/2023: Corn, silage

Crop: Corn, silage Acres planted: 50 Plant date: 05/24/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/16/2023	1,718.54 *ton*	As-is		67.0	2,700.00	700.00	3,700.00		5.09

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	34.37	185.60	48.12	254.34	1,154.65

NUTRIENT BUDGET

A. LAND APPLICATIONS

1 - 10/28/2022: Oats, silage-soft dough

Field name: 1

Crop: Oats, silage-soft dough Plant date: 10/28/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/07/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater sludge	65.66	7.71	84.90	246.99	702,942.00 *gal*
Application event totals		65.66	7.71	84.90	246.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/04/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	84.43	7,209,000.00 *gal*
Application event totals		0.00	0.00	0.00	84.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/19/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	64.24	5.59	113.98	783.83	792,000.00 *gal*
Application event totals		64.24	5.59	113.98	783.83	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	108.76	11.54	194.80	854.95	924,000.00 *gal*
Application event totals		108.76	11.54	194.80	854.95	

1 - 05/24/2023: Corn, silage

Field name: 1

Crop: Corn, silage Plant date: 05/24/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

1 - 05/24/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWQ	Process wastewater	105.52	9.19	156.53	746.73	792,000.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	3.47	640,800.00 *gal*
Application event totals		105.52	9.19	156.53	750.20	

05/24/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	10.00	0.00	0.00	0.00	
Application event totals		10.00	0.00	0.00	0.00	

06/20/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	40.50	7,476,000.00 *gal*
Application event totals		0.00	0.00	0.00	40.50	

07/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	74.34	5.47	145.57	644.50	759,000.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	33.27	6,141,000.00 *gal*
Application event totals		74.34	5.47	145.57	677.76	

07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	40.50	7,476,000.00 *gal*
Application event totals		0.00	0.00	0.00	40.50	

07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	40.50	7,476,000.00 *gal*
Application event totals		0.00	0.00	0.00	40.50	

1 - 05/24/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	71.11	5.23	139.24	616.48	726,000.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	31.82	5,874,000.00 *gal*
Application event totals		71.11	5.23	139.24	648.29	

08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	37.60	6,942,000.00 *gal*
Application event totals		0.00	0.00	0.00	37.60	

2 - 10/28/2022: Oats, silage-soft dough

Field name: 2

Crop: Oats, silage-soft dough Plant date: 10/28/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/07/2022	Towed tank	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater sludge	65.60	7.70	84.83	246.78	308,045.00 *gal*
Application event totals		65.60	7.70	84.83	246.78	

10/04/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	85.56	3,204,000.00 *gal*
Application event totals		0.00	0.00	0.00	85.56	

01/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	168.27	17.85	301.38	1,322.73	627,000.00 *gal*
Application event totals		168.27	17.85	301.38	1,322.73	

2 - 05/24/2023: Corn, silage

Field name: 2

Crop: Corn, silage

Plant date: 05/24/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/03/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	193.07	98.12	437.84	4,557.77	420.00 *ton*
Application event totals		193.07	98.12	437.84	4,557.77	

05/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	42.87	3,471,000.00 *gal*
Application event totals		0.00	0.00	0.00	42.87	

05/24/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	10.00	0.00	0.00	0.00	
Application event totals		10.00	0.00	0.00	0.00	

06/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	39.57	3,204,000.00 *gal*
Application event totals		0.00	0.00	0.00	39.57	

07/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	39.57	3,204,000.00 *gal*
Application event totals		0.00	0.00	0.00	39.57	

2 - 05/24/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	39.57	3,204,000.00 *gal*
Application event totals		0.00	0.00	0.00	39.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	37.92	3,070,500.00 *gal*
Application event totals		0.00	0.00	0.00	37.92	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	66.32	4.88	129.87	575.00	297,000.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	29.68	2,403,000.00 *gal*
Application event totals		66.32	4.88	129.87	604.68	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	36.27	2,937,000.00 *gal*
Application event totals		0.00	0.00	0.00	36.27	

3 - 10/28/2022: Oats, silage-soft dough

Field name: 3

Crop: Oats, silage-soft dough

Plant date: 10/28/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/07/2022	Towed tank	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater sludge	65.60	7.70	84.83	246.78	234,114.00 *gal*
Application event totals		65.60	7.70	84.83	246.78	

3 - 10/28/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/03/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	163.14	17.31	292.20	1,282.42	462,000.00 *gal*
Application event totals		163.14	17.31	292.20	1,282.42	

10/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	84.43	2,403,000.00 *gal*
Application event totals		0.00	0.00	0.00	84.43	

3 - 05/24/2023: Corn, silage

Field name: 3

Crop: Corn, silage

Plant date: 05/24/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/03/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	217.75	110.66	493.81	5,140.35	360.00 *ton*
Application event totals		217.75	110.66	493.81	5,140.35	

05/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	41.22	2,536,500.00 *gal*
Application event totals		0.00	0.00	0.00	41.22	

05/24/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	10.00	0.00	0.00	0.00	
Application event totals		10.00	0.00	0.00	0.00	

3 - 05/24/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	39.05	2,403,000.00 *gal*
Application event totals		0.00	0.00	0.00	39.05	

07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	39.05	2,403,000.00 *gal*
Application event totals		0.00	0.00	0.00	39.05	

07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	39.05	2,403,000.00 *gal*
Application event totals		0.00	0.00	0.00	39.05	

08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	39.05	2,403,000.00 *gal*
Application event totals		0.00	0.00	0.00	39.05	

08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	67.88	4.99	132.91	588.45	231,000.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	30.37	1,869,000.00 *gal*
Application event totals		67.88	4.99	132.91	618.83	

08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	36.88	2,269,500.00 *gal*
Application event totals		0.00	0.00	0.00	36.88	

9 - 10/28/2022: Oats, silage-soft dough

Field name: 9

Crop: Oats, silage-soft dough — Plant date: 10/28/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/24/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	67.13	5.84	119.11	819.11	726,000.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	78.43	5,874,000.00 *gal*
Application event totals		67.13	5.84	119.11	897.53	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	128.42	13.62	230.00	1,009.45	957,000.00 *gal*
Application event totals		128.42	13.62	230.00	1,009.45	

9 - 05/24/2023: Corn, silage

Field name: 9

Crop: Corn, silage — Plant date: 05/24/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	100.24	8.74	148.71	709.39	660,000.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	32.98	5,340,000.00 *gal*
Application event totals		100.24	8.74	148.71	742.37	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/24/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	10.00	0.00	0.00	0.00	
Application event totals		10.00	0.00	0.00	0.00	

9 - 05/24/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	41.22	6,675,000.00 *gal*
Application event totals		0.00	0.00	0.00	41.22	

07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	70.01	5.15	137.09	606.95	627,000.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	31.33	5,073,000.00 *gal*
Application event totals		70.01	5.15	137.09	638.28	

07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	41.22	6,675,000.00 *gal*
Application event totals		0.00	0.00	0.00	41.22	

08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	70.01	5.15	137.09	606.95	627,000.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	31.33	5,073,000.00 *gal*
Application event totals		70.01	5.15	137.09	638.28	

08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	39.57	6,408,000.00 *gal*
Application event totals		0.00	0.00	0.00	39.57	

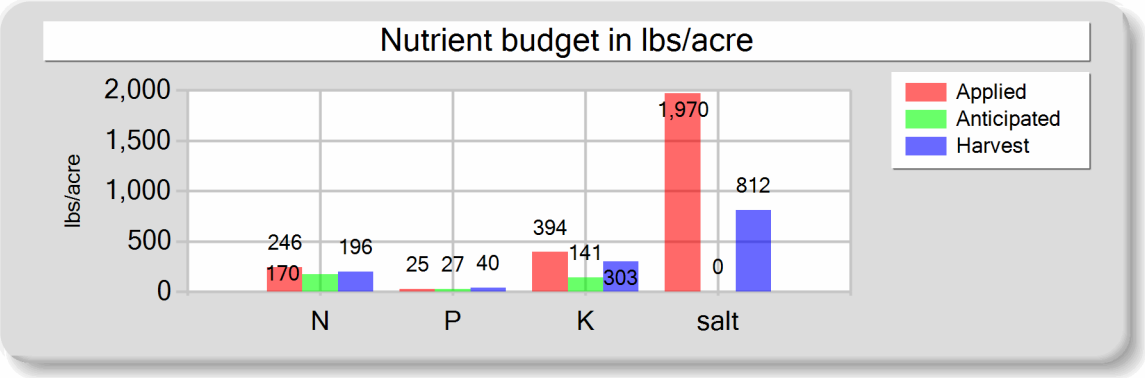
08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	37.92	6,141,000.00 *gal*
Application event totals		0.00	0.00	0.00	37.92	

B. NUTRIENT BUDGET

1 - 10/28/2022: Oats, silage-soft dough

Field name: 1 Crop: Oats, silage-soft dough Plant date: 10/28/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	238.66	24.84	393.68	1,885.77
Fresh water	0.00	0.00	0.00	84.43
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	245.66	24.84	393.68	1,970.21
Anticipated crop nutrient removal	170.00	27.20	141.10	0.00
Actual crop nutrient removal	196.30	39.93	302.77	812.03
Nutrient balance	49.36	-15.09	90.91	1,158.17
Applied to removed ratio	1.25	0.62	1.30	2.43

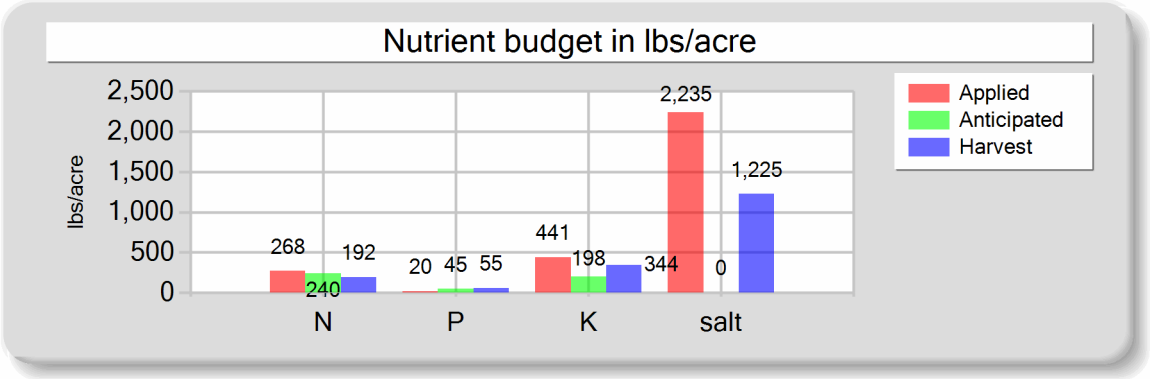
Fresh water applied	
7,209,000.00	*gallons*
265.48	*acre-inches*
4.66	*inches/acre*

Process wastewater applied	
2,418,942.00	*gallons*
89.08	*acre-inches*
1.56	*inches/acre*

Total harvests for the crop	
1	*harvests*

1 - 05/24/2023: Corn, silage

Field name: 1 Crop: Corn, silage Plant date: 05/24/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	10.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	250.96	19.89	441.34	2,007.70
Fresh water	0.00	0.00	0.00	227.65
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	267.96	19.89	441.34	2,235.35
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	192.48	54.99	343.71	1,224.97
Nutrient balance	75.49	-35.10	97.63	1,010.38
Applied to removed ratio	1.39	0.36	1.28	1.82

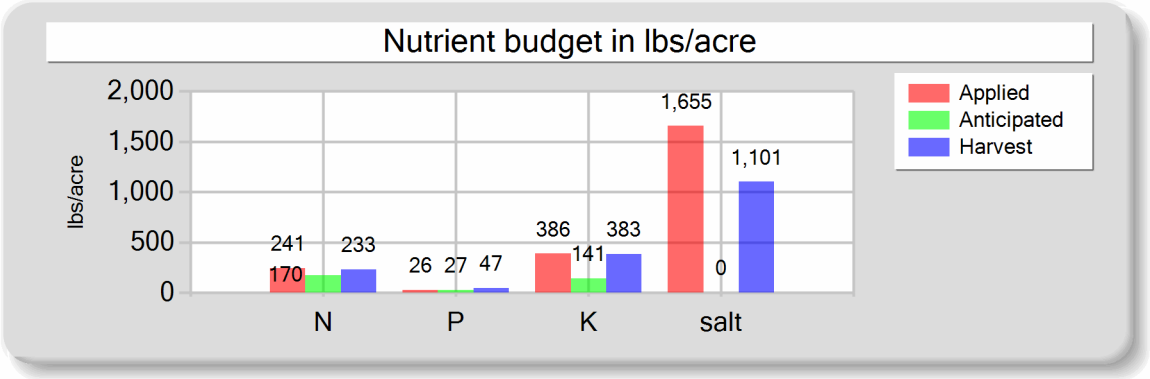
Fresh water applied	
42,025,800.00	*gallons*
1,547.67	*acre-inches*
27.15	*inches/acre*

Process wastewater applied	
2,277,000.00	*gallons*
83.85	*acre-inches*
1.47	*inches/acre*

Total harvests for the crop	
1	*harvests*

2 - 10/28/2022: Oats, silage-soft dough

Field name: 2 Crop: Oats, silage-soft dough Plant date: 10/28/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	233.87	25.55	386.21	1,569.51
Fresh water	0.00	0.00	0.00	85.56
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	240.87	25.55	386.21	1,655.07
Anticipated crop nutrient removal	170.00	27.20	141.10	0.00
Actual crop nutrient removal	232.90	46.58	382.62	1,100.68
Nutrient balance	7.98	-21.03	3.59	554.39
Applied to removed ratio	1.03	0.55	1.01	1.50

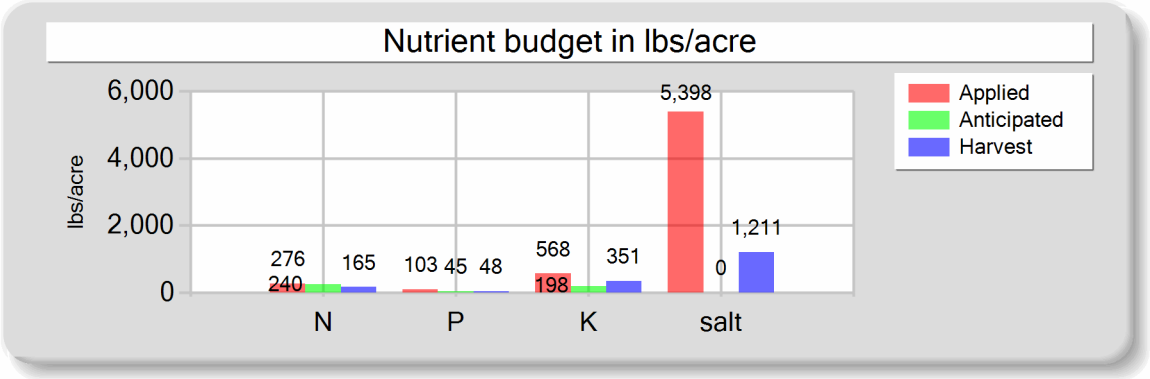
Fresh water applied	
3,204,000.00	*gallons*
117.99	*acre-inches*
4.72	*inches/acre*

Process wastewater applied	
935,045.00	*gallons*
34.43	*acre-inches*
1.38	*inches/acre*

Total harvests for the crop	
1	*harvests*

2 - 05/24/2023: Corn, silage

Field name: 2 Crop: Corn, silage Plant date: 05/24/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	10.00	0.00	0.00	0.00
Dry manure	193.07	98.12	437.84	4,557.77
Process wastewater	66.32	4.88	129.87	575.00
Fresh water	0.00	0.00	0.00	265.46
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	276.40	103.00	567.71	5,398.23
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	164.98	48.12	350.58	1,210.75
Nutrient balance	111.42	54.88	217.13	4,187.49
Applied to removed ratio	1.68	2.14	1.62	4.46

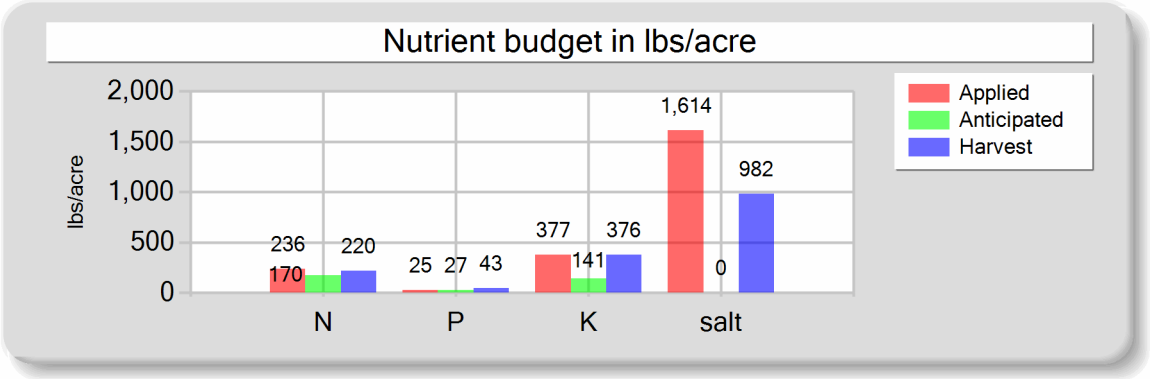
Fresh water applied	
21,493,500.00	*gallons*
791.53	*acre-inches*
31.66	*inches/acre*

Process wastewater applied	
297,000.00	*gallons*
10.94	*acre-inches*
0.44	*inches/acre*

Total harvests for the crop	
1	*harvests*

3 - 10/28/2022: Oats, silage-soft dough

Field name: 3 Crop: Oats, silage-soft dough Plant date: 10/28/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	228.75	25.01	377.03	1,529.21
Fresh water	0.00	0.00	0.00	84.43
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	235.75	25.01	377.03	1,613.64
Anticipated crop nutrient removal	170.00	27.20	141.10	0.00
Actual crop nutrient removal	219.59	43.25	375.96	981.65
Nutrient balance	16.16	-18.24	1.07	631.99
Applied to removed ratio	1.07	0.58	1.00	1.64

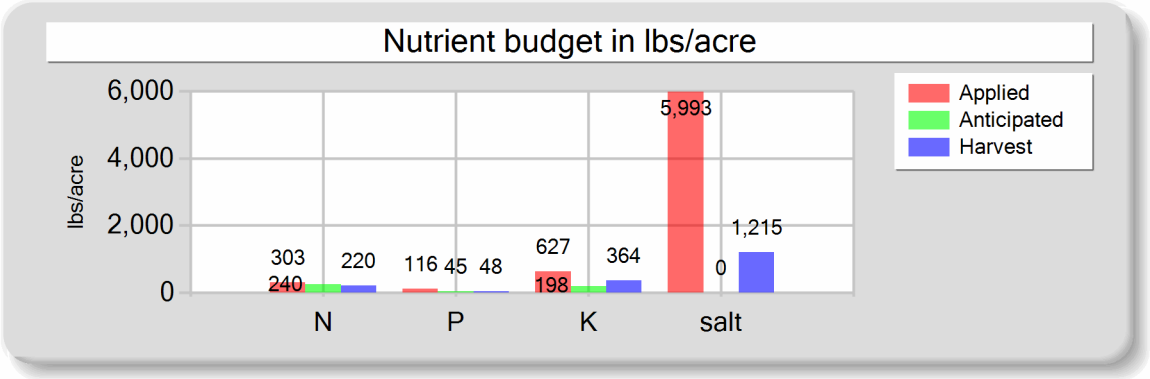
Fresh water applied	
2,403,000.00	*gallons*
88.49	*acre-inches*
4.66	*inches/acre*

Process wastewater applied	
696,114.00	*gallons*
25.64	*acre-inches*
1.35	*inches/acre*

Total harvests for the crop	
1	*harvests*

3 - 05/24/2023: Corn, silage

Field name: 3 Crop: Corn, silage Plant date: 05/24/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	10.00	0.00	0.00	0.00
Dry manure	217.75	110.66	493.81	5,140.35
Process wastewater	67.88	4.99	132.91	588.45
Fresh water	0.00	0.00	0.00	264.68
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	302.63	115.65	626.72	5,993.48
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	219.97	48.12	364.33	1,214.75
Nutrient balance	82.65	67.53	262.39	4,778.73
Applied to removed ratio	1.38	2.40	1.72	4.93

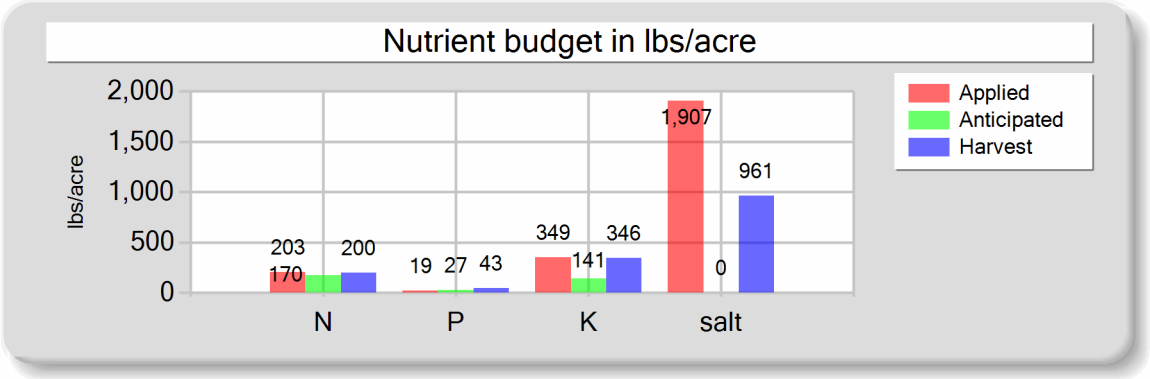
Fresh water applied	
16,287,000.00	*gallons*
599.79	*acre-inches*
31.57	*inches/acre*

Process wastewater applied	
231,000.00	*gallons*
8.51	*acre-inches*
0.45	*inches/acre*

Total harvests for the crop	
1	*harvests*

9 - 10/28/2022: Oats, silage-soft dough

Field name: 9 Crop: Oats, silage-soft dough Plant date: 10/28/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	195.55	19.46	349.11	1,828.56
Fresh water	0.00	0.00	0.00	78.43
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	202.55	19.46	349.11	1,906.99
Anticipated crop nutrient removal	170.00	27.20	141.10	0.00
Actual crop nutrient removal	199.63	43.25	346.02	961.20
Nutrient balance	2.92	-23.79	3.09	945.78
Applied to removed ratio	1.01	0.45	1.01	1.98

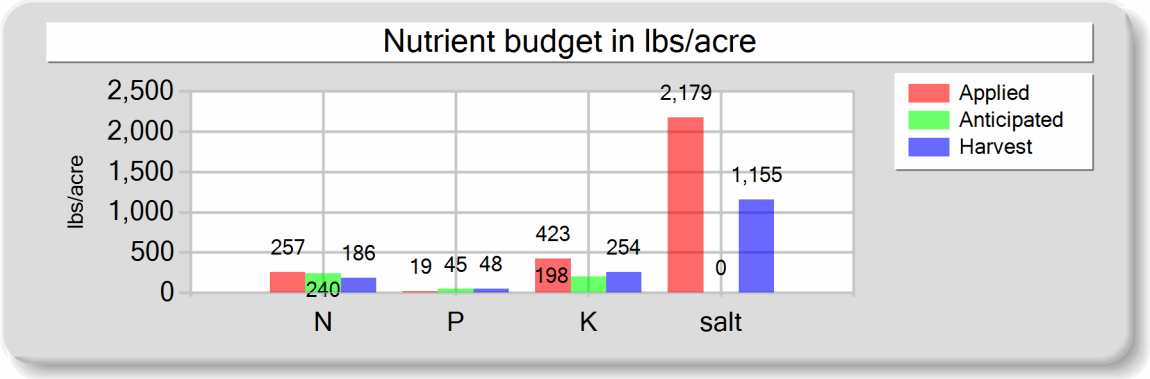
Fresh water applied	
5,874,000.00	*gallons*
216.32	*acre-inches*
4.33	*inches/acre*

Process wastewater applied	
1,683,000.00	*gallons*
61.98	*acre-inches*
1.24	*inches/acre*

Total harvests for the crop	
1	*harvests*

9 - 05/24/2023: Corn, silage

Field name: 9 Crop: Corn, silage Plant date: 05/24/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	10.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	240.26	19.03	422.88	1,923.29
Fresh water	0.00	0.00	0.00	255.56
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	257.26	19.03	422.88	2,178.85
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	185.60	48.12	254.34	1,154.65
Nutrient balance	71.65	-29.09	168.54	1,024.20
Applied to removed ratio	1.39	0.40	1.66	1.89

Fresh water applied	
41,385,000.00	*gallons*
1,524.07	*acre-inches*
30.48	*inches/acre*

Process wastewater applied	
1,914,000.00	*gallons*
70.49	*acre-inches*
1.41	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Corral Manure

Sample and source description: Corral Manure

Sample date: 04/21/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,300.00	9,300.00	41,500.00	0.00	0.00	0.00	0.00	0.00		43.20
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

Separator Manure

Sample and source description: Separator Manure

Sample date: 04/21/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 46.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,000.00	9,000.00	32,000.00	0.00	0.00	0.00	0.00	0.00		33.90
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

Corral Manure

Sample and source description: Corral Manure

Sample date: 09/18/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 19.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	28,600.00	9,220.00	30,400.00	21,600.00	8,000.00	9,220.00	6,240.00	2,890.00		26.90
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

Separator Manure

Sample and source description: Separator Manure

Sample date: 09/18/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 63.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	25,300.00	14,900.00	7,610.00	0.00	0.00	0.00	0.00	0.00		30.70
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

B. PROCESS WASTEWATER ANALYSES

Slurry

Sample and source description: Slurry

Sample date: 09/09/2022 Material type: Process wastewater sludge Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	638.00	353.00	0.00	0.00	74.90	825.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5,600.00	2,400
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

WWQ4

Sample and source description: WWQ4

Sample date: 10/20/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	554.00	300.00	0.00	0.00	48.20	983.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8,400.00	6,760
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

WWQ1

Sample and source description: WWQ1

Sample date: 02/10/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	804.00	507.00	0.00	0.00	85.30	1,440.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	10,200.00	6,320
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

WWQ2

Sample and source description: WWQ2

Sample date: 04/21/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	910.00	538.00	0.00	0.00	79.30	1,350.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	10,700.00	6,440
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

WWQ3

Sample and source description: WWQ3

Sample date: 07/20/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	669.00	294.00	0.00	0.00	49.20	1,310.00	187.00	144.00	450.00	4,260.00	0.00	460.00	466.00	9,440.00	5,800
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

Slurry

Sample and source description: Slurry

Sample date: 09/18/2023 Material type: Process wastewater sludge Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	851.00	61.60	0.00	0.00	206.00	969.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5,830.00	2,040
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

WWQ4

Sample and source description: WWQ4

Sample date: 10/16/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	627.00	431.00	0.00	0.00	45.60	1,190.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8,460.00	6,020
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

C. FRESH WATER ANALYSES

MID Lateral 5

MID Lateral 5

Sample description: MID Lateral 5

Sample date: 07/22/2022 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	228.00	80
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

MID Lateral 5

Sample description: MID Lateral 5

Sample date: 07/25/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	38.10	37
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	10.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

1 - 10/28/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,600.00	1,200.00	9,200.00		8.83
DL	0.05	0.02	0.02		0.05

Forage

Sample and source description: Forage

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 73.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,900.00	1,200.00	9,100.00		9.21
DL	0.05	0.02	0.02		0.05

1 - 05/24/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/26/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	2,500.00	700.00	5,100.00		6.17
DL	0.05	0.02	0.02		0.05

1 - 05/24/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/26/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	2,800.00	800.00	5,000.00		5.94
DL	0.05	0.02	0.02		0.05

2 - 10/28/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 72.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,000.00	1,400.00	11,500.00		11.90
DL	0.05	0.02	0.02		0.05

2 - 05/24/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/26/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	2,400.00	700.00	5,100.00		5.70
DL	0.05	0.02	0.02		0.05

3 - 10/28/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,600.00	1,300.00	11,300.00		10.50
DL	0.05	0.02	0.02		0.05

3 - 05/24/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/26/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,200.00	700.00	5,300.00		5.93
DL	0.05	0.02	0.02		0.05

9 - 10/28/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,900.00	1,200.00	10,200.00		9.62
DL	0.05	0.02	0.02		0.05

9 - 10/28/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 73.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,000.00	1,300.00	10,400.00		10.70
DL	0.05	0.02	0.02		0.05

9 - 05/24/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/26/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 72.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	2,200.00	800.00	4,500.00		6.10
DL	0.05	0.02	0.20		0.05

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/26/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	2,700.00	700.00	3,700.00		5.09
DL	0.05	0.02	0.02		0.05

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

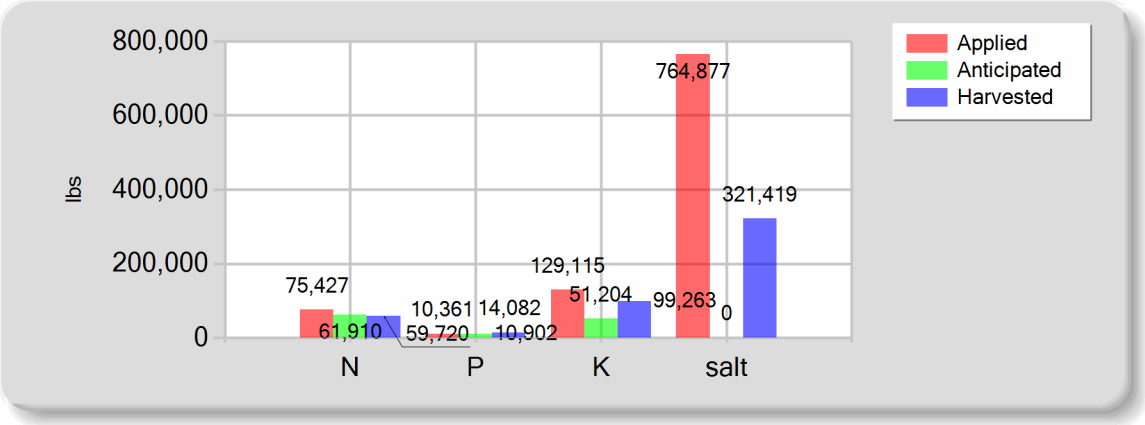
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

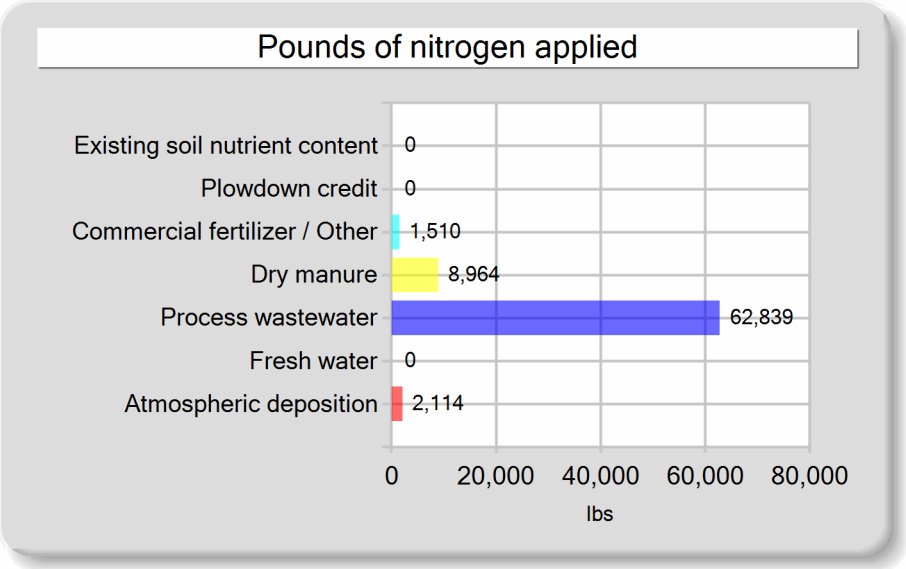
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

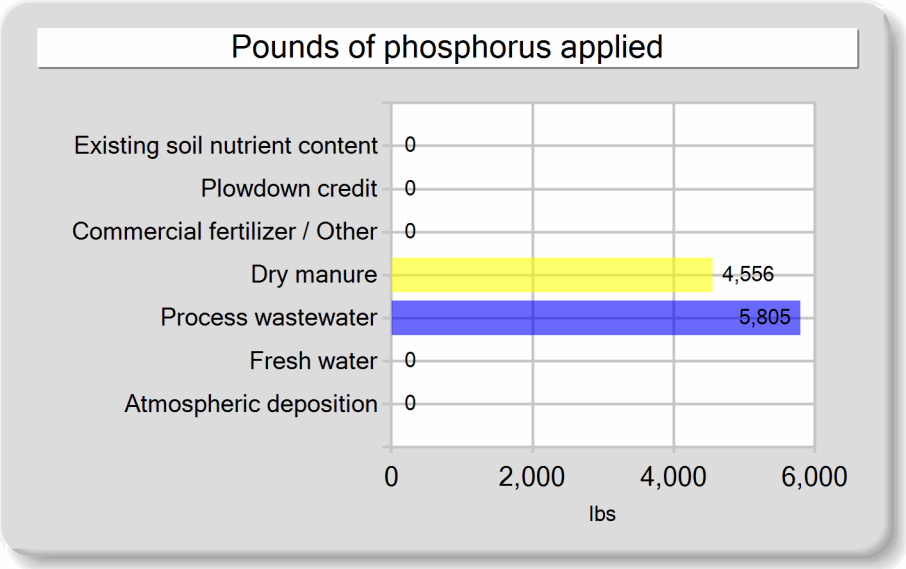
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	1,510.00	0.00	0.00	0.00
Dry manure	8,964.07	4,555.51	20,328.36	211,610.88
Process wastewater	62,839.26	5,805.17	108,786.86	503,368.75
Fresh water	0.00	0.00	0.00	49,897.08
Atmospheric deposition	2,114.00	0.00	0.00	0.00
Total nutrients applied	75,427.34	10,360.68	129,115.22	764,876.71
Anticipated crop nutrient removal	61,910.00	10,902.20	51,204.10	0.00
Actual crop nutrient removal	59,720.25	14,082.45	99,262.74	321,419.37
Nutrient balance	15,707.08	-3,721.77	29,852.48	443,457.34
Applied to removed ratio	1.26	0.74	1.30	2.38

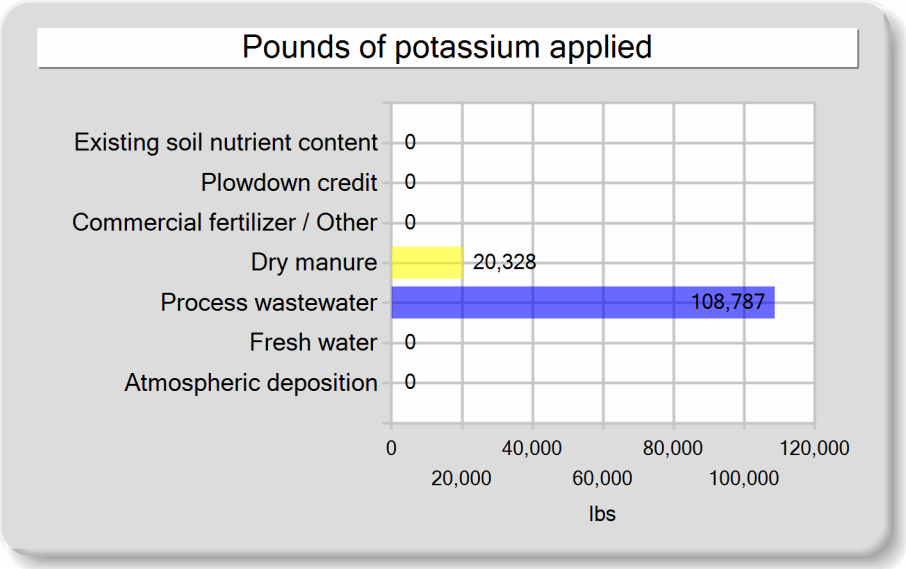
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

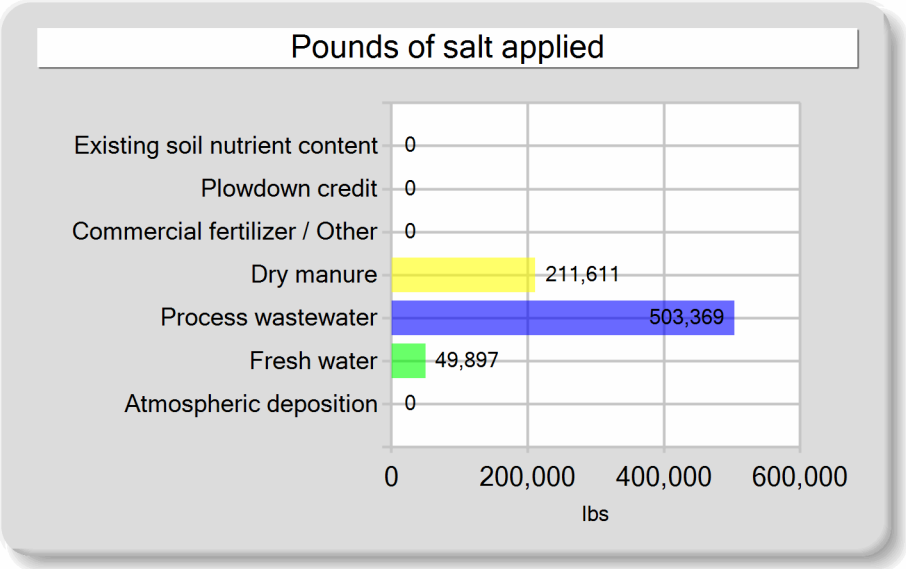


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

Field 2 Corn (1.68) had a high nitrogen field ratio as crop failed to meet anticipated nitrogen removal of 240 pounds/acre. Yield for the subject field was adequate however a low nitrogen value from the lab plant tissue analysis resulted in only 4.8 pounds of nitrogen removed per harvested ton compared to the anticipated 8 pounds of nitrogen removed per harvested ton which resulted in the high nitrogen field ratio. Nitrogen application of 269 pounds per acre was not considered excessive but in line with that recommended in the dairy's Nutrient Management Plan.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

jAMES D. mEIKLE jR
jAMES D. mEIKLE jR (Mar 26, 2024 14:12 PDT)

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
James D Meikle, Jr	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Mar 26, 2024	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Meikle Dairy
1685 S. Hart Rd.
Modesto, CA 95358

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 11/14/23 12:58 PM
SUBMITTED BY: Vince
ANALYZED DATE: 11/16/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 16.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW#1-1601 Hart	T3318035A	Electrical Conductivity (EC)	EPA 120.1 and	< 0.100	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	2.51	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	49.9	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	94.4	98.8 (T3317024A)	96.4 (T3317024A)	2.45
Electrical Conductivity (EC)	97.0			3.03
Nitrate Nitrogen (NO3-N)	99.5	98.9 (T3317005A)	98.7 (T3317005A)	0.223

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

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www.denelelabs.com

Compliance Analysis Report

Customer
Meikle Dairy
1685 S. Hart Rd.
Modesto, CA 95358

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 11/14/23 12:58 PM
SUBMITTED BY: Vince
ANALYZED DATE: 11/16/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 13.8c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW#3-Cows-Dairy	T3318035B	Electrical Conductivity (EC)	EPA 120.1 and	0.985	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	12.3	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	630	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	94.4	98.8 (T3317024A)	96.4 (T3317024A)	2.45
Electrical Conductivity (EC)	97.0			3.03
Nitrate Nitrogen (NO3-N)	99.5	98.9 (T3317005A)	98.7 (T3317005A)	0.223

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Compliance Analysis Report

Customer
Meikle Dairy
1685 S. Hart Rd.
Modesto, CA 95358

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 11/14/23 12:58 PM
SUBMITTED BY: Vince
ANALYZED DATE: 11/16/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 13.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW#4-Ranch	T3318035C	Electrical Conductivity (EC)	EPA 120.1 and	1.15	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	34.7	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	739	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	94.4	98.8 (T3317024A)	96.4 (T3317024A)	2.45
Electrical Conductivity (EC)	97.0			3.03
Nitrate Nitrogen (NO3-N)	99.5	98.9 (T3317005A)	98.7 (T3317005A)	0.223

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
Meikle Dairy
1685 S. Hart Rd.
Modesto, CA 95358

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 11/14/23 12:58 PM
SUBMITTED BY: Vince
ANALYZED DATE: 11/16/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 14.2c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW#5-Canal	T3318035D	Electrical Conductivity (EC)	EPA 120.1 and	1.40	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	59.0	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	897	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	94.4	98.8 (T3317024A)	96.4 (T3317024A)	2.45
Electrical Conductivity (EC)	97.0			3.03
Nitrate Nitrogen (NO3-N)	99.5	98.9 (T3317005A)	98.7 (T3317005A)	0.223

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Compliance Analysis Report

Customer
Meikle Dairy
1685 S. Hart Rd.
Modesto, CA 95358

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 11/14/23 12:58 PM
SUBMITTED BY: Vince
ANALYZED DATE: 11/15/23

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 14.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Irrigation Well	T3318035E	Electrical Conductivity (EC)	EPA 120.1 and	0.929	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	27.5	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	535	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	27.5	mg/L
		Soluble Salts (SALT-SOL)	Calculation	595	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	94.4	98.8 (T3317024A)	96.4 (T3317024A)			2.45
Electrical Conductivity (EC)	97.0			1.17 (T3317024A)	1.21 (Dup)	3.03
Nitrate Nitrogen (NO3-N)	99.5	98.9 (T3317005A)	98.7 (T3317005A)			0.223
Total Dissolved Solids (TDS)	102			50.0 (T3313017B)	50.0 (Dup)	0.000
Total Nitrogen (TN)	90.7			0.000 (T3313017B)	0.000 (null)	

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DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380

PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: Meikle Dairy
Mailing address: S. Hart Rd.
(not facility address) Modesto, CA 95358
Phone#:

Authorized Copy Released To: F+R Ag

Sampled By: Jim Meikle
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
H1 EC*, NO3-N, NH4-N*
H6 EC, TN, TDS
H7 Combination of H1 & H6
H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
H3 EC*, NH4-N, TKN, TP, K, TDS
H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
H8 Combination of H3 &H5
H9 Combination of H4 &H5

TILE DRAIN
H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
M1 % Moisture
M3 TN TP, K, % Moisture
M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
M7 Combination of M3 & M4

PLANT TISSUE
F18 % Moisture, Ash, TN, P, K, DM and as received
F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
K3 Soluble Phosphorus
K4 1ft NO3, Organic Matter
K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
K7 Combination of K3 & K4
K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	11-14-23	7:30	DW #1	1601 Hart	16.0°C		✓				
~~2~~	~~11-14-23~~	~~8:00~~	~~DW #2~~	~~1685 Hart~~			~~✓~~				
3	11-14-23	8:10	DW #3	Cows- Dairy	13.8°		✓				
4	11-14-23	8:25	DW #4	Ranch	13.0°		✓				
5	11-14-23	8:45	DW #5	Canal	14.2°		✓				
6	11-14-23	9:15	Irrigation Well		14.0°						H7
7											
8											
9											
10											

Analysis

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	Logged In By:	LABORATORY NUMBER
1st	Vince Furtado	F+R	11-14-23 11:00	11-14-23		
2nd		DENELE ANALYTICAL, INC.	11-14-23 12:58			T3318035

(LABORATORY USE ONLY)

Required to be submitted to any agency? (YES / NO)

Denele Analytical, Inc.

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Compliance Analysis Report

Customer
Meikle Dairy
1685 S. Hart Rd.
Modesto, CA 95358

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 11/29/23 12:45 PM
SUBMITTED BY: Vince
ANALYZED DATE: 11/30/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 15.2c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW #2 - 1685 Hart	T3333011A	Electrical Conductivity (EC)	EPA 120.1 and	1.13	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	32.6	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	721	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	95.6			
Electrical Conductivity (EC)	96.0			0.032
Nitrate Nitrogen (NO3-N)	93.8	102 (T3333002A)	98.3 (T3333002A)	3.56

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ELAP Certificate No.: 2714

DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380

PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: Meikle Dairy
Mailing address: S. Hart Road
(not facility address) Modesto, CA 95358
Phone#:

Authorized Copy Released To: F+R Ag

Sampled By: Vince Furtado
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	11-29-23	7:30	DW #2 - 1685 Hart		15.2°		✓				
2											
3											
4											
5											
6											
7											
8											
9											
10											

Analysis

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	Logged In By:	LABORATORY NUMBER
1st	Vince Furtado	F+R Ag		11-29-23		
2nd		DENELE ANALYTICAL, INC.	11-29-23 12:45			T3333011

(LABORATORY USE ONLY)

Required to be submitted to any agency? (YES / NO)

2023 Annual Report Meikle Ranch

Final Audit Report 2024-03-26

Created:	2024-03-26
By:	Vince Furtado (vfurtado@fragservices.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAAF_RCueWKVCLmJ0qHK602nxRsNHBIYNxY

"2023 Annual Report Meikle Ranch" History

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Signer jme1070622@aol.com entered name at signing as jAMES D. mEIKLE jR
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